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TECHNICAL SPECIFICATION

Core Network and Interoperability Testing (INT);
5G NAS Conformance Testing for the N1 interface;
(3GPP™ Release 16);
Part 2: Test Suite Structure (TSS) and Test Purposes (TP)

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Core Network and Interoperability Testing (INT).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [2].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Test Suite Structure (TSS) and Test Purposes (TP) for the test specification for the 5G NAS protocol on the N1 interface as specified in ETSI TS 124 501 [1] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.2] and ETSI ETS 300 406 [i.3].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 124 501 (V16.15.0): "5G; Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3 (3GPP TS 24.501 version 16.15.0 Release 16)".

[2] ETSI TS 103 921-1: "Core Network and Interoperability Testing (INT); 5G NAS Conformance Testing for the N1 interface; (3GPPTM Release 16); Part 1: Protocol Implementation Conformance Statement (PICS)".

[3] ETSI TS 123 502 (V16.19.0): "5G; Procedures for the 5G System (5GS) (3GPP TS 23.502 version 16.19.0 Release 16) ".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ISO/IEC 9646-1: "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 1: General concepts".

[i.2] ISO/IEC 9646-7: "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 7: Implementation Conformance Statements".

[i.3] ETSI ETS 300 406: "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TS 124 501 [1] and the following apply:

Abstract Test Method (ATM): Refer to ISO/IEC 9646-1 [i.1].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [i.1].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [i.1].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [i.1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TS 124 501 [1] and the following apply:

TS	Test System
TSS	Test Suite Structure

4 Test configurations

4.1 Introduction

Test purposes of the present document address the 5G functional entities UE and AMF that are accessible via the standardized N1 interface.

4.2 Test configuration using the N2/N1 interface

The N1 interface is located between the UE and the AMF. Following configurations are simplified to highlight tested interface and involved entities. Overall network infrastructure is shown under clause 4.3.

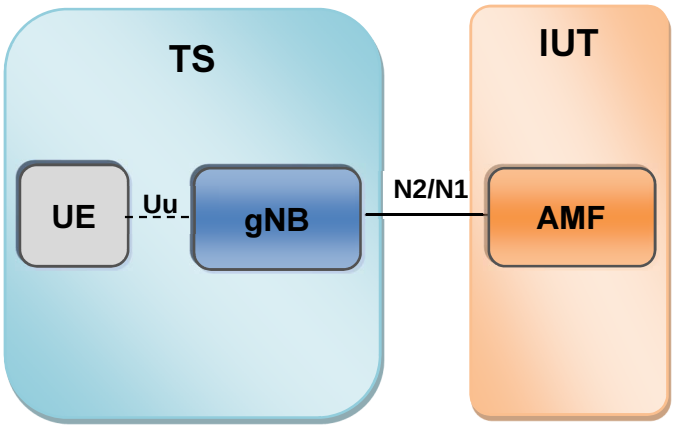


Figure 1: Test configuration CF_AMF_N2N1

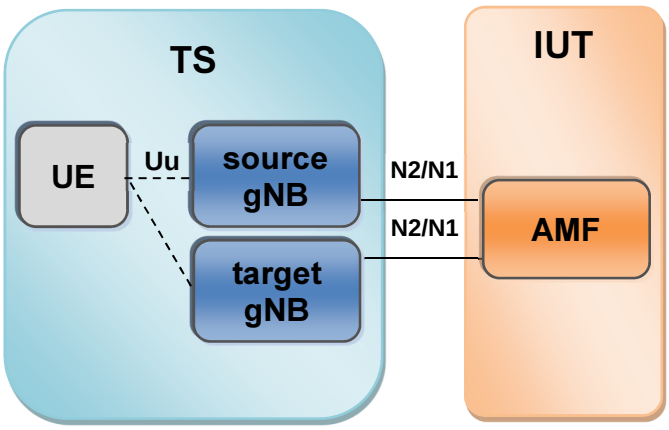


Figure 2: Test configuration CF_AMF_2N2N1

4.3 Network infrastructure

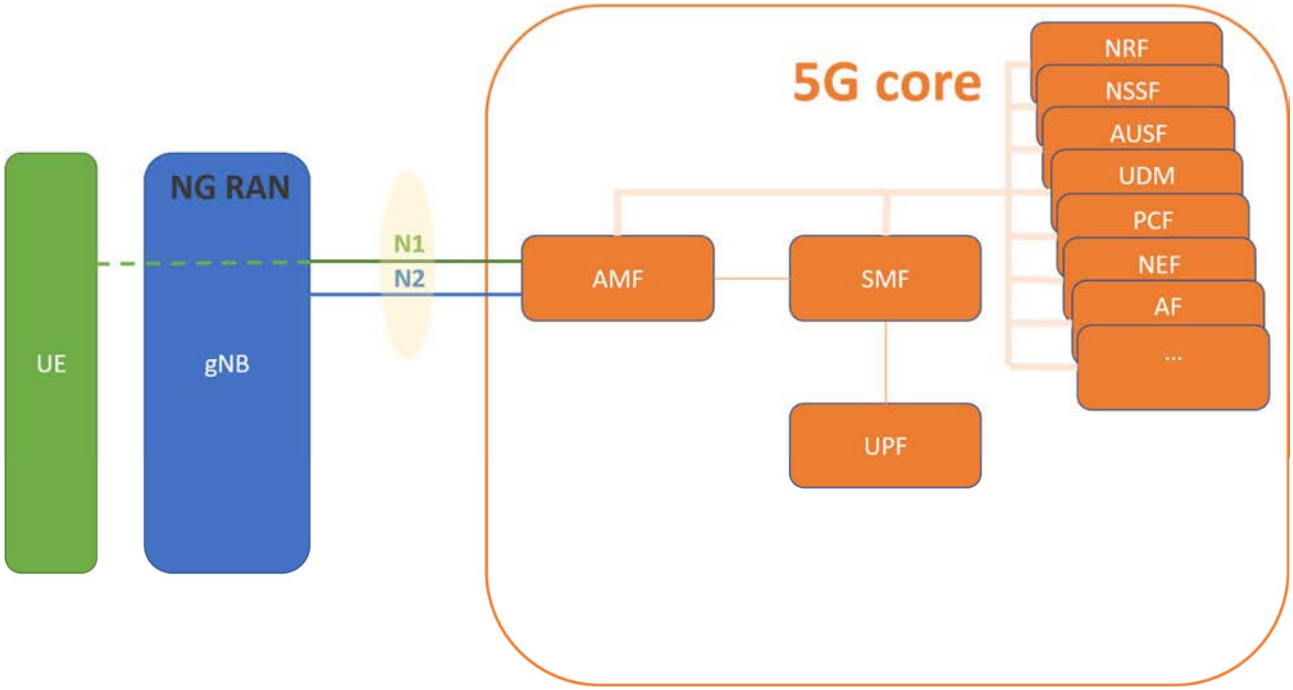


Figure 3: Network architecture

5 Test Suite Structure (TSS) and Test Purposes (TP)

5.1 Test Suite Structure

5.1.1 TP naming convention

TPs are numbered, starting at 01, within each group. Groups are organized according to the TSS.

Table 1: TP identifier naming convention scheme

Identifier: **<TP>_<iut>_<scope1>_<scope2>_<nn>**			
<tp>	=	Test Purpose:	fixed to "TP"
<interface or protocol>			Interface or protocol: 5GNAS
<iut>	=	type of IUT:	AMF
<scope1>	=	group1	AUT 5GMM / Authentication procedure
			SEC 5GMM / Security mode control procedure
			IDN 5GMM / Identification procedure
			CON 5GMM / Generic UE configuration update procedure
			ULN 5GMM / Uplink NAS transport procedure
			DLN 5GMM / Downlink NAS transport procedure
			EST 5GMM / EMM Status
			NSA 5GMM / Network slice-specific authentication and authorization procedure
			REG 5GMM / Registration procedure
			DRG 5GMM / De-registration procedure
			SRV 5GMM / Service request procedure
			PAG 5GMM / Paging procedure
			NOT 5GMM / Notification procedure
			NPE 5GSM / Network-requested PDU session establishment procedure
			NPM 5GSM / Network-requested PDU session modification procedure
			NPR 5GSM / Network-requested PDU session release procedure
			UPA 5GSM / Network-requested PDU session authentication and authorization procedure
			UPE 5GSM / UE requested PDU session establishment procedure
			UPM 5GSM / UE requested PDU session modification procedure
			UPR 5GSM / UE requested PDU session release procedure
			SST 5GSM / Status procedure
<scope2>	=	group2	REQ Request
			REJ Reject
			ACC Accept
			RES Response
			ABN Abnormal Case
			COM Command
<nn>	=	sequential number	(01 to 99)

5.1.2 Test strategy

As the base specification in ETSI TS 124 501 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification ETSI TS 103 921-1 [2].

5.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used which is illustrated in Table 2. Table 2 should be read in conjunction with any TP, i.e. each individual TP can be used as an example to facilitate the full comprehension of Table 2.

Table 2: Structure of a single TP

TP part	Text	Example
Header	**<Identifier>**	**see Table 1**
	<clause number in base ETSI TS 124 501 [1]>	**clause 8.2.6**
	<PICS reference>	**A.4/3**
Summary	*Short free text description of the test objective*	Verify that the IUT can successfully process all mandatory IEs in a REGISTRATION REQUEST received due to registration procedure
Configuration	*Test configuration as described in clause 4.2*	CF_AMF_N2N1
Initial condition (optional)	*Free text description of the condition that the IUT has reached before the test purpose applies.*	
Start point	Ensure that the IUT in the	Having sent a REGISTRATION REQUEST
	<state> *see ETSI TS 124 501 [1], clause 8.1*	
	and/or further actions before stimulus *if the action is sending/receiving* *see below for message structure*	
Stimulus	<**trigger**>, *see below for message structure*	on receipt of a REGISTRATION REQUEST (see note 2)
	or <goal>	
Reaction	<**action**>.	sends, saves, does, etc.
	if the action is sending *see below for message structure* <next action>, *etc.*	
Message structure	<message type>	Message exchange, etc. (see note 2)
	a) containing a(n) <IE name> IE (see note 4) b) indicating <coding of the field> and *back to a) or b)* (see note 3)	

NOTE 1: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.
NOTE 2: All messages are considered as "valid and compatible" unless otherwise specified in the test purpose. This includes the presence of all mandatory IEs as specified in ETSI TS 124 501 [1].
NOTE 3: An IE can be embedded into another IE. This is expressed by indentations, e.g. if Message1 contains IE1 and IE2 where IE1 has IE3 embedded this will be expressed like this:

```
sends/receives Message 1
    containing IE1
        containing IE3
            indicating ...
    containing IE2
        indicating ...
```

5.2 Test Purposes

5.2.0 Introduction

The present document contains modified and new Test Purposes. Some Test Purpose Identifiers may not be backward compatible with previous versions of the specification.

5.2.1 PICS references

All PICS items referred to in this clause are as specified in ETSI TS 103 921-1 [2] unless indicated otherwise by another numbered reference. PICS items are only meant for test selection, therefore only PICS items with status optional or conditional are explicitly mentioned.

5.2.2 N1 interface 5G NAS - AMF Role

5.2.2.1 Test selection

The IUT takes the role of the AMF; PICS A.2/1.

5.2.2.2 Events for 5G NAS - AMF Role

Events used in the Test Purposes for the 5G NAS AMF role are described in Table 3.

Table 3: Events defined for 5G NAS AMF Role

Event	Description
indicateUEdeactivation	The UE is administratively deactivated in the 5GC.
indicateUEderegistration	The UE is administratively de-registered in the 5GC.
isInOverloadedState	The AMF is in an overload condition and overload control procedures are active.

5.2.2.3 Authentication procedure

TP Id	TP_5GNAS_AMF_AUT_REQ_01
Test Objective	Verify that the IUT sends an AUTHENTICATION_REQUEST message upon receipt of an initial REGISTRATION_REQUEST containing a SUCI, when no active 5G NAS security context exists.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.2 and 8.2.1
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_1_2

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
        NAS_PDU containing {
            extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
            security_header_type set to NOT_SECURITY_PROTECTED,
            message_type set to REGISTRATION_REQUEST,
            5GS_registration_type containing {
                5GS_registration_type_value set to INITIAL_REGISTRATION},
            5GS_mobile_identity containing {
                SUPI_format set to IMSI,
                Type_of_identity set to SUCI,
                MCC set to PX_MCC,
                MNC set to PX_MNC,
                Routing_indicator set to PX_ROUTING_INDICATOR,
                Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                Home_network_public_key_identifier set to PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                MSIN set to PX_MSIN}}}
    }
    then {
        the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_REQUEST,
                ngKSI containing {
                    nas_key_set_identifier set to "111"},
                ABBA containing {
                    ABBA_contents set to " '0000'H "},
                Authentication_parameter_RAND set to RV_Authentication_parameter_RAND,
```

```
                Authentication_parameter_AUTN set to RV_Authentication_parameter_AUTN}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_AUT_REQ_02
Test Objective	Verify that the IUT sends an AUTHENTICATION_REJECT message upon receipt of an AUTHENTICATION_RESPONSE containing an invalid authentication response value, when the UE was identified with SUCI in the initial NAS message.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.5 and 8.2.5
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_1_2

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    the IUT receives an INITIAL_UE_MESSAGE containing {
        NAS_PDU containing {
            extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
            security_header_type set to NOT_SECURITY_PROTECTED,
            message_type set to REGISTRATION_REQUEST,
            5GS_registration_type containing {
                5GS_registration_type_value set to INITIAL_REGISTRATION},
            5GS_mobile_identity containing {
                Type_of_identity set to SUCI}}} and
    event AUTHENTICATION_REQUEST occurs
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_RESPONSE,
                authentication_response_parameter containing {
                    RES set to PX_INVALID_RES_VALUE}}}
        from the GNB entity
    }
    then {
        the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_REJECT}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_AUT_REQ_03
Test Objective	Verify that the IUT stops re-sending an AUTHENTICATION_REQUEST message if no AUTHENTICATION_RESPONSE message is received on the fifth expire of timer T3560.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.7 b) and Table 10.3.1
Configuration	CF_AMF_N2N1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
    and (.) at time point start_initial_T3560 : event AUTHENTICATION_REQUEST occurs
    and (!) 6s after @start_initial_T3560 : event AUTHENTICATION_REQUEST occurs
    and (!) 12s after @start_initial_T3560 : event AUTHENTICATION_REQUEST occurs
    and (!) 18s after @start_initial_T3560 : event AUTHENTICATION_REQUEST occurs
}
```

Expected Behaviour

```
ensure that {
    when {
        (.) at time point start_trigger_T3560 : event AUTHENTICATION_REQUEST occurs
    }
    then {
        (!) 6s after @start_trigger_T3560 : the IUT aborts an AUTHENTICATION_REQUEST procedure
        //(Note 1: "It is not expected to receive another AUTHENTICATION message on the N2N1
interface.")
    }
}
```

TP Id	TP_5GNAS_AMF_AUT_REQ_04
Test Objective	Verify that the IUT sends an IDENTITY_REQUEST message correctly upon receipt of an AUTHENTICATION_FAILURE message indicating a 5GMM cause value #20 - MAC failure.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.7 c), 5.4.3 and 8.2.4
Configuration	CF_AMF_N2N1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    event AUTHENTICATION_REQUEST occurs
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED,
                message_type set to AUTHENTICATION_FAILURE,
                5GMM_cause set to MAC_failure}}
        from the GNB entity
    }
    then {
        the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED,
                message_type set to IDENTITY_REQUEST,
                identity_type set to SUCI}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_AUT_REQ_05
Test Objective	Verify that the IUT sends a new AUTHENTICATION_REQUEST message with new ngKSI value to re-initiate the 5G AKA based primary authentication upon receipt of an AUTHENTICATION_FAILURE message indicating a 5GMM cause value #71 - ngKSI already in use.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.7 e) and 8.2.4
Configuration	CF_AMF_N2N1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_REQUEST,
                ngKSI containing {
                    nas_key_set_identifier set to "111"},
                ABBA containing {
                    ABBA_contents set to " '0000'H "},
                Authentication_parameter_RAND set to RV_Authentication_parameter_RAND,
                Authentication_parameter_AUTN set to RV_Authentication_parameter_AUTN}}
    to the GNB entity
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED,
                message_type set to AUTHENTICATION_FAILURE,
                5GMM_cause set to ngKSI_already_in_use}}
        from the GNB entity
    }
    then {
        the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_REQUEST,
```

```
                ngKSI containing {
                    nas_key_set_identifier set to PX_NON_DEFAULT_NGKSI},
                ABBA containing {
                    ABBA_contents set to " '0000'H "},
                Authentication_parameter_RAND set to RV_Authentication_parameter_RAND,
                Authentication_parameter_AUTN set to RV_Authentication_parameter_AUTN}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_AUT_ABN_01
Test Objective	Verify that the IUT sends an IDENTITY_REQUEST message to obtain the SUCI from the UE upon receipt of an AUTHENTICATION_FAILURE message indicating a 5GMM cause value #26 - non-5G authentication unacceptable.
Reference	ETSI TS 124 501 [1], clauses 5.4.1.3.7, 8.2.4, 5.4.3.2 and 8.2.21.1
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_1_2_3_1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_REQUEST,
                ngKSI containing {
                    nas_key_set_identifier set to "111"},
                ABBA containing {
                    ABBA_contents set to " '0000'H "},
                Authentication_parameter_RAND set to RV_Authentication_parameter_RAND,
                Authentication_parameter_AUTN set to
PX_Authentication_parameter_AUTN_NON_5G_UNACCEPTABLE}}
    to the GNB entity
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED,
                message_type set to AUTHENTICATION_FAILURE,
                5GMM_cause set to "Non-5G authentication unacceptable (26)"}}
        from the GNB entity
    }
    then {
        the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
                NAS_PDU containing {
                    extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                    security_header_type set to NOT_SECURITY_PROTECTED,
                    message_type set to IDENTITY_REQUEST,
                    identity_type set to SUCI}}
        to the GNB entity
    }
}
```

5.2.2.4 Security mode control procedure

TP Id	TP_5GNAS_AMF_SEC_COM_01
Test Objective	Verify that the IUT sends a SECURITY_MODE_COMMAND message to initiate the NAS security mode control procedure upon receipt of an AUTHENTICATION_RESPONSE message in a 5G AKA based primary authentication procedure.
Reference	ETSI TS 124 501 [1], clauses 5.4.2.2, 5.4.2.3 and 8.2.25
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_2_1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    event AUTHENTICATION_REQUEST occurs with {
        argument replaced by the AUTHENTICATION_REQUEST containing {
            security_header_type set to NOT_SECURITY_PROTECTED}
```

```
    }
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to AUTHENTICATION_RESPONSE}}
    from the GNB entity
    }
    then {
    the IUT sends an DOWNLINK_NAS_TRANSPORT containing {
        NAS_PDU containing {
            extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
            security_header_type set to INTEGRITY_PROTECTED_WITH_NEW_5G_NAS_SECURITY_CONTEXT,
            message_type set to SECURITY_MODE_COMMAND,
            selected_NAS_security_algorithms containing {
                type_of_ciphering_algorithm set to PX_SELECTED_NAS_CIPHERING_ALGORITHM,
                type_of_integrity_algorithm set to PX_SELECTED_NAS_INTEGRITY_ALGORITHM},
            ngKSI containing {
                nas_key_set_identifier set to "111"},
            replayed_UE_security_capabilities set to PX_UE_SECURITY_CAPABILITIES}}
    to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_SEC_ACC_01
Test Objective	Verify that the IUT, upon receiving the NAS SECURITY_MODE_COMPLETE message after completing the NAS Authentication and Security procedure, successfully completes the registration process by accepting the registration.
Reference	ETSI TS 124 501 [1], clauses 5.4.2.4, 5.5.1.2.4, 8.2.7 and 8.2.26
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_2

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to
INTEGRITY_PROTECTED_AND_CIPHERED_WITH_NEW_5G_NAS_SECURITY_CONTEXT,
                message_type set to SECURITY_MODE_COMPLETE}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
           NAS_PDU containing {
               extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
               security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
               message_type set to REGISTRATION_ACCEPT,
               5GS_registration_result containing {
                   5GS_registration_result_value set to 3GPP_ACCESS},
               5G_GUTI containing {
                   Type_of_identity set to 5G_GUTI,
                   MCC set to PX_MCC,
                   MNC set to PX_MNC,
                   AMF_Region_ID set to PX_AMF_REGION_ID,
                   AMF_Set_ID set to PX_AMF_SET_ID,
                   AMF_Pointer set to PX_AMF_POINTER,
                   5G_TMSI set to RV_5G_TMSI},
               TAI_list containing {
                   Partial_tracking_area_list_1 containing {
                       MCC set to PX_MCC,
                       MNC set to PX_MNC,
                       TAC set to PX_TAC}},
               T3512_value containing {
                   Timer_value set to nonZeroValue}}}
        to the GNB entity
```

```
    }
}
```

TP Id	TP_5GNAS_AMF_SEC_REJ_01
Test Objective	Verify that the IUT aborts the ongoing procedure that triggered the NAS security mode control procedure upon receipt of a SECURITY_MODE_REJECT message containing 5GMM cause #23 'UE security capabilities mismatch'.
Reference	ETSI TS 124 501 [1], clauses 5.4.2.5 and 8.2.27
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_2

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF and
    (.) at time point start_SMC_event : event SECURITY_MODE_COMMAND occurs
}
```

Expected Behaviour

```
ensure that {
    when {
        (.) at time point start_SMC_event : event SECURITY_MODE_COMMAND occurs and
        the IUT receives a UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_WITH_NEW_5G_NAS_SECURITY_CONTEXT,
                message_type set to SECURITY_MODE_REJECT,
                5GMM_cause set to "UE security capabilities mismatch (23)"}}
        from the GNB entity
    }
    then {
        // PX_SEC_ABORT_TWAIT: observation timer to verify that no further registration
progression occurs
        (!) PX_SEC_ABORT_TWAIT after @start_SMC_event : the IUT does not send a
INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                message_type set to REGISTRATION_ACCEPT}}
        to the GNB entity
        //(Note 1: "The IUT is expected to abort the ongoing procedure and shall not proceed with
registration completion.")
    }
}
```

5.2.2.5 Identification procedure

TP Id	TP_5GNAS_AMF_IDN_REQ_01
Test Objective	Verify that the IUT initiates the identification procedure by sending an IDENTITY_REQUEST message with identity type set to SUCI when the IUT receives a REGISTRATION_REQUEST message containing a 5G-GUTI that is not known by the IUT.
Reference	ETSI TS 123 502 [3], clause 4.2.2.2.2 ETSI TS 124 501 [1], clauses 5.4.3.2 and 8.2.21
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_3

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_mobile_identity containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to PX_UNASSIGNED_5G_TMSI}}}
```

```
        from the GNB entity
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                message_type set to IDENTITY_REQUEST,
                identity_type set to SUCI}}
        to the GNB entity
    }
}
```

5.2.2.6 Generic UE configuration update procedure

TP Id	TP_5GNAS_AMF_CNF_COM_01
Test Objective	Verify that the IUT does not retransmit a CONFIGURATION_UPDATE_COMMAND containing only NITZ-related information elements when the Configuration update indication IE is absent or indicates that acknowledgement is not requested.
Reference	ETSI TS 124 501 [1], clauses 5.4.4.2 and 8.2.19
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_4_1
Initial Conditions	

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        (.) at time point start_CUC_event :the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to CONFIGURATION_UPDATE_COMMAND,
                Full_name_for_network,
                Short_name_for_network,
                Local_time_zone,
                Universal_time_and_local_time_zone,
                Daylight_saving_time,
                not Configuration_update_indication OR
                Configuration_update_indication containing {
                    Acknowledgement_requested set to 0
                }}}
        to the GNB entity
    }
     then {
         // PX_CUC_TWAIT value for timer awaiting no response after CUC message
         (!) PX_CUC_TWAIT after @start_CUC_event : the IUT does not send a DOWNLINK_NAS_TRANSPORT
containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to CONFIGURATION_UPDATE_COMMAND}}
        to the GNB entity
        //(Note 1: "It is not expected to receive a retransmission of the same NITZ-only
CONFIGURATION_UPDATE_COMMAND on the N2N1 interface.")
     }
}
```

5.2.2.7 NAS transport procedure

TP Id	TP_5GNAS_AMF_DLN_ACC_01
Test Objective	Verify that the IUT, upon receipt of a UL NAS TRANSPORT message containing a PDU SESSION ESTABLISHMENT REQUEST, sends a PDU_SESSION_RESOURCE_SETUP_REQUEST carrying a DL NAS TRANSPORT message with a PDU SESSION ESTABLISHMENT ACCEPT.
Reference	ETSI TS 124 501 [1], clauses 5.4.5, 6.4.1, 8.2.10 and 8.2.11
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A5_6
Initial Conditions	

```
with {
 the UE isRegisteredToAMF
}
```

Expected Behaviour

ensure that {
when {
the IUT receives an UPLINK_NAS_TRANSPORT containing {
NAS_PDU containing {
extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
message_type set to UL_NAS_TRANSPORT,
payload_container_type set to "N1 SM information",
payload_container containing {
5GSM_PDU containing {
extended_protocol_discriminator set to 5GS_SESSION_MANAGEMENT_MESSAGES,
message_type set to PDU_SESSION_ESTABLISHMENT_REQUEST,
PDU_session_type set to "IPv4",
SSC_mode set to "SSC mode 1"}},
pDU_session_identity set to PX_PDU_session_identity,
request_type set to "Initial request",
s_NSSAI containing {
sst set to PX_SST,
sd set to PX_SD},
dNN set to PX_DNN}}
from the GNB entity
}
then {
the IUT sends a PDU_SESSION_RESOURCE_SETUP_REQUEST containing {
NAS_PDU containing {
extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
message_type set to DL_NAS_TRANSPORT,
payload_container_type set to "N1 SM information",
payload_container containing {
5GSM_PDU containing {
extended_protocol_discriminator set to 5GS_SESSION_MANAGEMENT_MESSAGES,
message_type set to PDU_SESSION_ESTABLISHMENT_ACCEPT}},
pDU_session_identity set to PX_PDU_session_identity}}
to the GNB entity
}
}

5.2.2.8 5GMM status procedure

Void.

5.2.2.9 Registration procedure

5.2.2.9.1 Registration procedure for initial registration

TP Id	TP_5GNAS_AMF_REG_ACC_01
Test Objective	Verify that, after successful completion of the initial registration procedure over 3GPP access, the IUT sends a REGISTRATION_ACCEPT message containing the 5GS registration result, TAI list, 5G-GUTI and T3512 value.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.4 and 8.2.7
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_8_1
Initial Conditions	

with {
the UE isNotRegisteredToAMF
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INITIAL_UE_MESSAGE containing {
NAS_PDU containing {
extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
security_header_type set to NOT_SECURITY_PROTECTED,
message_type set to REGISTRATION_REQUEST,
5GS_registration_type containing {
5GS_registration_type_value set to INITIAL_REGISTRATION},
5GS_mobile_identity containing {
SUPI_format set to IMSI,
Type_of_identity set to SUCI,	

```
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN}}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to REGISTRATION_ACCEPT,
                5GS_registration_result containing {
                    5GS_registration_result_value set to 3GPP_ACCESS},
                5G_GUTI containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI},
                TAI_list containing {
                    Partial_tracking_area_list_1 containing {
                        MCC set to PX_MCC,
                        MNC set to PX_MNC,
                        TAC set to PX_TAC}},
                T3512_value containing {
                    Timer_value set to nonZeroValue}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_ACC_02
Test Objective	Verify that, when the UE requests SMS over NAS transport during initial registration and the IUT supports SMS over NAS in initial registration, the IUT sends a REGISTRATION_ACCEPT message with the SMS allowed bit in the 5GS registration result IE set to 'SMS over NAS allowed'.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.4 and 8.2.7
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_8_1_3

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_registration_type containing {
                    5GS_registration_type_value set to INITIAL_REGISTRATION},
                5GS_mobile_identity containing {
                    SUPI_format set to IMSI,
                    Type_of_identity set to SUCI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN},
                5GS_update_type containing {
                    SMS_over_NAS_transport_requested set to 1}}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
```

```
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to REGISTRATION_ACCEPT,
                5GS_registration_result containing {
                    5GS_registration_result_value set to 3GPP_ACCESS,
                    SMS_over_NAS_transport_allowed set to 1}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_ACC_03
Test Objective	Verify that the IUT sends a REGISTRATION_ACCEPT message with the SMS over NAS transport allowed bit set to 'not allowed' when the UE requests SMS over NAS during initial registration and the IUT does not support SMS over NAS in initial registration.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.4 and 8.2.7
Configuration	CF_AMF_N2N1
PICS Selection	not PICS_A4_8_1_3

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_registration_type containing {
                    5GS_registration_type_value set to INITIAL_REGISTRATION},
                5GS_mobile_identity containing {
                    SUPI_format set to IMSI,
                    Type_of_identity set to SUCI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN},
                5GS_update_type containing {
                    SMS_over_NAS_transport_requested set to 1}}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to REGISTRATION_ACCEPT,
                5GS_registration_result containing {
                    5GS_registration_result_value set to 3GPP_ACCESS,
                    SMS_over_NAS_transport_allowed set to 0}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_ACC_04
Test Objective	Verify that the IUT, during the initial registration procedure, sends a REGISTRATION_ACCEPT message including the Allowed NSSAI IE when the UE includes a Requested NSSAI IE in the REGISTRATION_REQUEST message and the IUT allows one or more S-NSSAIs from the requested NSSAI.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.4 and 8.2.7
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_8_1

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_registration_type containing {
                    5GS_registration_type_value set to INITIAL_REGISTRATION},
                5GS_mobile_identity containing {
                    SUPI_format set to IMSI,
                    Type_of_identity set to SUCI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN},
                Requested_NSSAI containing {
                    S_NSSAI_1 containing {
                        SST set to PX_SST_1,
                        SD set to PX_SD_1},
                    S_NSSAI_2 containing {
                        SST set to PX_SST_2,
                        SD set to PX_SD_2}}}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                  security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to REGISTRATION_ACCEPT,
                5G_GUTI containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI},
                Allowed_NSSAI containing {
                    S_NSSAI_1 containing {
                        SST set to PX_SST_1,
                        SD set to PX_SD_1}}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_ACC_05
Test Objective	Verify that, during the initial registration procedure, the IUT includes the Rejected NSSAI IE in the REGISTRATION_ACCEPT message when the UE includes a Requested NSSAI IE in the REGISTRATION_REQUEST message and the IUT rejects one or more S-NSSAIs from the requested NSSAI.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.4 and 8.2.7
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_8_1_2_3

Initial Conditions

```
with {
    the UE isNotRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_registration_type containing {
                    5GS_registration_type_value set to INITIAL_REGISTRATION},
                5GS_mobile_identity containing {
                    SUPI_format set to IMSI,
                    Type_of_identity set to SUCI,
```

```
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN},
                Requested_NSSAI containing {
                    S_NSSAI_1 containing {
                        SST set to PX_SST_1,
                        SD set to PX_SD_1},
                    S_NSSAI_2 containing {
                        SST set to PX_SST_Rejected,
                        SD set to PX_SD_Rejected}}}}
        from the GNB entity
    }
    then {
        the IUT sends a INITIAL_CONTEXT_SETUP_REQUEST containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                 security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to REGISTRATION_ACCEPT,
                5G_GUTI containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI},
                Rejected_NSSAI containing {
                    S_NSSAI_1 containing {
                        SST set to PX_SST_Rejected,
                        SD set to PX_SD_Rejected}}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_REJ_01
Test Objective	Verify that the IUT sends a REGISTRATION_REJECT message with 5GMM cause value #22 'congestion' and includes the T3346 value IE when the initial registration request is rejected due to general NAS level mobility management congestion control.
Reference	ETSI TS 124 501 [1], clauses 5.5.1.2.5 and 8.2.9
Configuration	CF_AMF_N2N1
Initial Conditions	

```
with {
    the UE isNotRegisteredToAMF and
    the AMF isInOverloadedState
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to NOT_SECURITY_PROTECTED,
                message_type set to REGISTRATION_REQUEST,
                5GS_registration_type containing {
                    5GS_registration_type_value set to INITIAL_REGISTRATION},
                5GS_mobile_identity containing {
                    SUPI_format set to IMSI,
                    Type_of_identity set to SUCI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    Routing_indicator set to PX_ROUTING_INDICATOR,
                    Protection_scheme_id set to PX_PROTECTION_SCHEME_ID,
                    Home_network_public_key_identifier set to
PX_HOME_NETWORK_PUBLIC_KEY_IDENTIFIER,
                    MSIN set to PX_MSIN}}}
        from the GNB entity
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
```

```
                message_type set to REGISTRATION_REJECT,
                5GMM_cause set to "Congestion (22)",
                T3346_value containing {
                    Timer_value set to nonZeroValue}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_REG_REJ_02
Test Objective	Verify that, during the initial registration procedure, the IUT sends a REGISTRATION_REJECT message with 5GMM cause value #9 'UE identity cannot be derived by the network' when the UE responds to an IDENTITY_REQUEST for SUCI with an IDENTITY_RESPONSE indicating that no identity is available.
Reference	ETSI TS 123 502 [3], clause 4.2.2.2.2 ETSI TS 124 501 [1], clauses 5.4.3.2, 5.4.3.3, 5.5.1.2.5 and 8.2.9
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_3
Initial Conditions	

```
with {
    the UE isNotRegisteredToAMF and
    the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
        NAS_PDU containing {
            extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
            message_type set to IDENTITY_REQUEST,
            identity_type set to SUCI}}
    to the GNB entity
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                message_type set to IDENTITY_RESPONSE,
                5GS_mobile_identity containing {
                    Type_of_identity set to NO_IDENTITY}}}
        from the GNB entity
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                message_type set to REGISTRATION_REJECT,
                5GMM_cause set to "UE identity cannot be derived by the network (9)"}}
        to the GNB entity
    }
}
```

5.2.2.9.2 Registration procedure for mobility and periodic registration update

Void.

5.2.2.10 De-registration procedure

5.2.2.10.1 UE initiated de-registration

TP Id	TP_5GNAS_AMF_DRG_ACC_01
Test Objective	Verify that the IUT sends a DEREGISTRATION_ACCEPT message upon receipt of a UE-originated DEREGISTRATION_REQUEST message indicating normal de-registration.
Reference	ETSI TS 124 501 [1], clauses 5.5.2.2.3, 8.2.12 and 8.2.13
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_9_1
Initial Conditions	

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST,
                Deregistration_type containing {
                    Switch_off set to NORMAL_DEREGISTRATION,
                    Reregistration_required set to REREGISTRATION_NOT_REQUIRED,
                    Access_type set to 3GPP_ACCESS},
                5GS_mobile_identity containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI}}}
        from the GNB entity
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_ACCEPT}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_DRG_ACC_02
Test Objective	Verify that the IUT does not send a DEREGISTRATION_ACCEPT message upon receipt of a UE-originated DEREGISTRATION_REQUEST message indicating switch off.
Reference	ETSI TS 124 501 [1], clauses 5.5.2.2.3, 8.2.12 and 8.2.13
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_9_1

Initial Conditions

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        (.) at time point start_DRG_event : the IUT receives an UPLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST,
                Deregistration_type containing {
                    Switch_off set to SWITCH_OFF,
                    Reregistration_required set to REREGISTRATION_NOT_REQUIRED,
                    Access_type set to 3GPP_ACCESS},
                5GS_mobile_identity containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI}}}
        from the GNB entity
    }
    then {
        // PX_DRG_TWAIT: Test system observation timer used to verify that no
DEREGISTRATION_ACCEPT message is sent after receipt of a switch-off DEREGISTRATION_REQUEST (no
corresponding NAS timer defined in the specification)
        (!) PX_DRG_TWAIT after @start_DRG_event : the IUT does not send a DOWNLINK_NAS_TRANSPORT
containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_ACCEPT}}
        to the GNB entity
```

```
    }
}
```

5.2.2.10.2 Network initiated de-registration

TP Id	TP_5GNAS_AMF_DRG_REQ_01
Test Objective	Verify that the IUT initiates network de-registration by sending a DEREGISTRATION_REQUEST message containing the De-registration type IE with re-registration not required for 3GPP access. NOTE: Explicit network deregistration triggered by O&M - deactivation of UE.
Reference	ETSI TS 124 501 [1], clauses 5.5.2.3.1 and 8.2.14
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_9_2

Initial Conditions

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT indicateUEdeactivation
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST,
                Deregistration_type containing {
                    Switch_off set to NORMAL_DEREGISTRATION,
                    Reregistration_required set to REREGISTRATION_NOT_REQUIRED,
                    Access_type set to 3GPP_ACCESS},
                5GS_mobile_identity containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI}}}
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_DRG_REQ_02
Test Objective	Verify that the IUT retransmits the DEREGISTRATION_REQUEST message if no DEREGISTRATION_ACCEPT is received from the UE. NOTE: Explicit network deregistration triggered by O&M - UE deregistration.
Reference	ETSI TS 124 501 [1], clauses 5.5.2.3.1 and 8.2.14
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_9_2

Initial Conditions

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT indicateUEdeactivation and
        (.) at time point start_DRG_event : the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST}}
        to the GNB entity
    }
    then {
        (!) PX_T3522 after @start_DRG_event : the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST}}
```

```
        to the GNB entity
    }
}
```

TP Id	TP_5GNAS_AMF_DRG_REQ_03
Test Objective	Verify that the IUT initiates network de-registration by sending a DEREGISTRATION_REQUEST message containing the De-registration type IE with re-registration required for 3GPP access.
Reference	ETSI TS 124 501 [1], clauses 5.5.2.3.1, 5.5.2.3.2 and 8.2.14
Configuration	CF_AMF_N2N1
PICS Selection	PICS_A4_9_2

Initial Conditions

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT indicateUEderegistration
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                extended_protocol_discriminator set to 5GS_MOBILITY_MANAGEMENT_MESSAGES,
                security_header_type set to INTEGRITY_PROTECTED_AND_CIPHERED,
                message_type set to DEREGISTRATION_REQUEST,
                Deregistration_type containing {
                    Switch_off set to NORMAL_DEREGISTRATION,
                    Reregistration_required set to REREGISTRATION_REQUIRED,
                    Access_type set to 3GPP_ACCESS},
                5GS_mobile_identity containing {
                    Type_of_identity set to 5G_GUTI,
                    MCC set to PX_MCC,
                    MNC set to PX_MNC,
                    AMF_Region_ID set to PX_AMF_REGION_ID,
                    AMF_Set_ID set to PX_AMF_SET_ID,
                    AMF_Pointer set to PX_AMF_POINTER,
                    5G_TMSI set to RV_5G_TMSI}}}
        to the GNB entity
    }
}
```

5.2.2.11 Service request procedure

Void.

5.2.2.12 Paging procedure

Void.

5.2.2.13 Notification procedure

Void.

5.2.2.14 UE-requested PDU session establishment procedure

Void.

5.2.2.15 PDU session authentication and authorization procedure

Void.

5.2.2.16 UE-requested PDU session modification procedure

Void.

5.2.2.17 Network-requested PDU session modification procedure

Void.

5.2.2.18 UE-requested PDU session release procedure

Void.

5.2.2.19 Network-requested PDU session release procedure

Void.

5.2.2.20 5GSM status procedure

Void.

Annex A (normative): Test Purpose source files

All TPs from clause 5 are defined in the file TP_AMF_NGNAS.tdltx in a machine-readable format. This file is available at the ETSI Forge repository:

https://forge.etsi.org/rep/int/5g-core/nas/-/tree/v1.2.1/test_purposes?ref_type=tags

History

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